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7.2.2 Nested dissection

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corresponding subvector of the right-hand side. And notice that that subject on the right-hand side is ordered similarly to the points that are part of this submatrix right here. So what does that mean? Hmm.

Well what that translates to is that this block of our matrix that corresponds to this right here, only captures how those values are connected to themselves. Okay? Now there's some

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Reading assignment

1/1 point (graded)

Read Unit 7.2.2 of the notes. [[LINK](#)]

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Homework 7.2.2.1

1/1 point (graded)
Consider the SPD matrix

$$A = \left(\begin{array}{c|c|c} A_{00} & 0 & A_{20}^T \\ \hline 0 & A_{11} & A_{21}^T \\ \hline A_{20} & A_{21} & A_{22} \end{array} \right).$$

- What special structure does the Cholesky factor of this matrix have?

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$$L = \left(\begin{array}{c|c|c} L_{00} & 0 & 0 \\ \hline 0 & L_{11} & 0 \\ \hline L_{20} & L_{21} & L_{22} \end{array} \right).$$

☐

$$L = \left(\begin{array}{c|c|c} L_{00} & 0 & 0 \\ \hline L_{10} & L_{11} & 0 \\ \hline L_{20} & L_{21} & L_{22} \end{array} \right).$$

☐

$$L = \left(\begin{array}{c|c|c} L_{00} & 0 & 0 \\ \hline L_{10} & L_{11} & 0 \\ \hline 0 & L_{21} & L_{22} \end{array} \right).$$



- How can the different parts of the Cholesky factor be computed in a way that takes advantage of the zero blocks?

☒ done



- How do you take advantage of the zero pattern when solving with the Cholesky factors?

☒ done



Solution:

For a complete solution see the [notes](#).

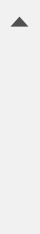
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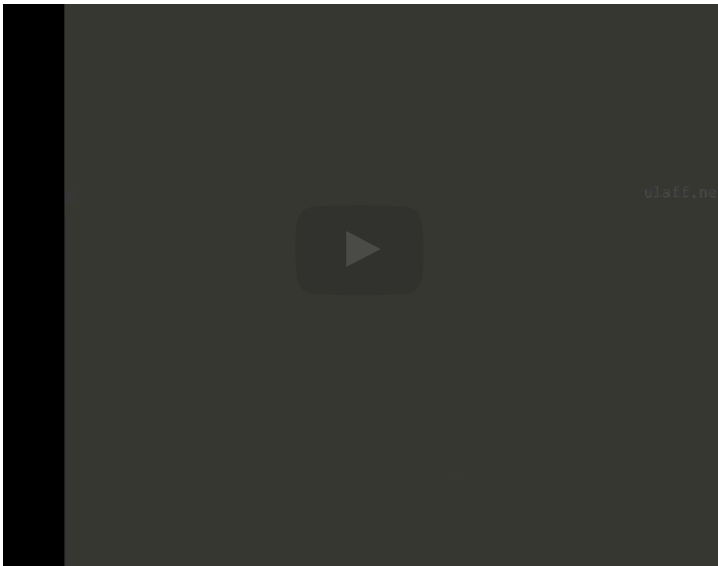
Answers are displayed within the problem

Video



carefully even, if you have a much more complex domain like this, what it really boils down to is trying to find separators





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that separate roughly
equally the graph into
two parts that are not
connected and to
then do that

recursively. Now, you
could say well that
means that we can
only do this if we

understand the graph
that underlies the
problem. But there is
this connection

between the nonzero
pattern in the matrix
and the graph that
describes that

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